From quality assurance to quality engineering: a multi-agent, local-first approach for agentic DevOps

Student: João Monteiro

Supervisor: Ilídio Oliveira

Co-supervisors: Sérgio Freire, Pedro Lobo

Summary: Software development, and quality engineering in particular, is undergoing a major shift: LLMs and agentic frameworks are making these practices increasingly automated. However, current cloud-based solutions present drawbacks for industrial adoption: high operational costs, data-privacy concerns and latency. This work addresses these limitations through local-first and hybrid solutions, focused specifically on Quality Engineering.

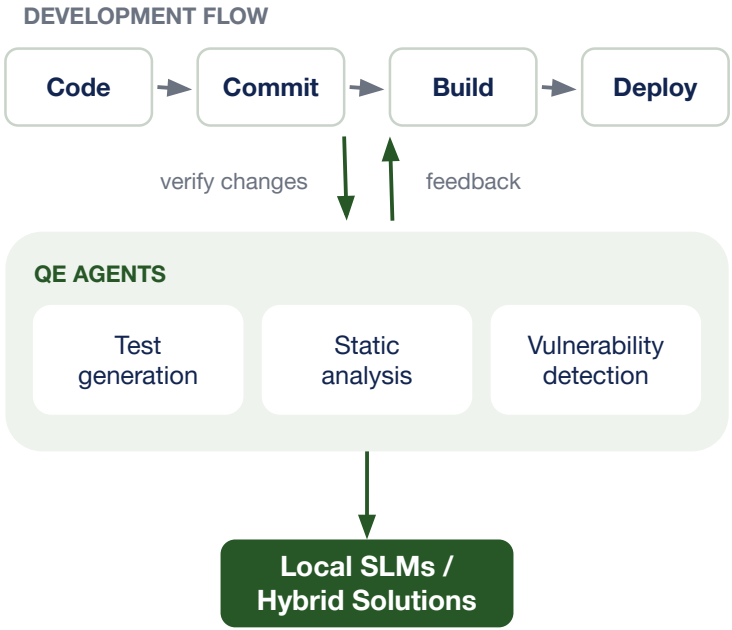
Objectives / Work done / Results

1. Multi-agent QE architecture
Specialised agents for test generation, static analysis and vulnerability detection

2. Local-first & hybrid strategies
Lower operational costs and address data-privacy concerns, with effectiveness comparable to cloud

3. Validation in real pipelines
Integrate the solution into real DevOps pipelines, with case studies from STIC systems

- **Literature review:** papers that survey current LLM-based solutions for Software Engineering, and that argue SLMs can be better suited than LLMs for agentic tasks
- First development of an **exploratory demo** to get familiar with the technologies and challenges ahead



Next Steps / Challenges / Bottlenecks

Next steps

- **Extend the literature review:** papers that evaluate current agentic solutions for Quality Engineering, and where SLMs outperform LLMs
- **Define research questions and evaluation metrics**
- **Start writing the state of the art**
- **Validate the current demo with the supervisors**

Challenge

Running local models on personal hardware: limited resources constrain which models can be tested, which matters for a local-first approach